Session plan 9: Large-N observational studies

Thu 22.10.26, 16:00–17:30 · Room 4.B51 · Basic reading: AJ ch. 2 (pp. 47–81); additional: KW pp. 92–99; Cinelli, Forney & Pearl (2024) · Applied reading: Ziblatt, Hilbig & Bischof (2024), *Wealth of tongues* · Presenters: **Ibrahima Khalil Ndiaye, Ella Schertenleib**

Learning goals: students can (1) explain why observational comparisons suffer from selection bias; (2) name two routes: natural experiments and control (compare like with like); (3) explain how the private-college premium shrinks when like is compared with like (AJ); (4) tell a good control (confounder) from bad controls (mediator, collider) in a DAG; (5) state when a left-out variable biases an estimate, and why robustness checks matter.

Before class: take-home exercise II uploaded to OLAT (beginner + advanced PDFs, materials folder; **not** the solutions) · presentation slides in OLAT (due Wed 21.10, 12:00)? · grading sheet `session09_handwritten.pdf` ready · session 8 board list ("why we can't just flip a coin") still visible or photographed · deck in speaker view (press S)

Checkpoints: 16:11 start "Compare like with like" · 16:20 start the three DAGs · **16:33 start pairs** · **16:52 hand over** · **16:53 presentation starts** (25 min max., then discussion until 17:27). About 6 min buffer before 16:52 (backup slide 17 if time)

Time	Slide	Content	Interaction	Guiding questions	If behind
16:00–16:03	2	**Bridge**: what can't be randomized (session 8 board list and pairs) → observational data (KW pp. 93–94)	Question	What is the main danger?	
16:03–16:07	3	**Is a top university worth it?** (AJ pp. 47–50)	Collect reasons on the board	Who goes there? Could we randomize it?	
16:07–16:11	4	**Two routes**: natural experiments (Oregon; Vietnam draft lottery, Erikson & Stoker 2011; thresholds) vs. control	Question	Which route is open for the university question?	2 min
16:11–16:16	5	**Compare like with like**: college matching matrix, groups A and B (AJ Table 2.1, pp. 51–55)	Students compute	Premium within A? Within B?	Give the answers
16:16–16:20	6	**The private premium shrinks**: 14% → 10% → 9% → about 1% (AJ Table 2.2, p. 63)	Guess before each reveal	Where does it end?	
16:20–16:23	7	**Good control: confounder** (family background; Cinelli Model 1; session 3 DAG 3)	Question	Which other confounders?	
16:23–16:26	8	**Bad control: mediator** (first job and network; Cinelli Model 11)	Question	What if we control for the first job?	
16:26–16:29	9	**Bad control: collider** (famous actors: talent and looks, hypothetical; Cinelli Models 16–17; session 6 admission example)	Question	Good-looking ones more or less talented?	
16:29–16:33	10	**What you leave out**: OVB in words, robustness (AJ pp. 68–79; formula and Table 2.5 in notes)	Question	What could still be missing?	2 min
16:33–16:43	11	**Pairs** on seminar questions 9.1–9.2: check DAGs; confounder, mediator or collider? what is missing?	6 min pairs, 4 min share		One pair shares
16:43–16:46	12	**Take-home exercise II** (Ziblatt et al., own controls; deadline 05.11.26); exercise I due 29.10.26	Admin		

Time	Slide	Content	Interaction	Guiding questions	If behind
16:52–16:53	13	Hand over			
16:53–17:18	—	**Student presentation** (25 min max.): summary and critique ~10 min, alternative design ~15 min	Presentation; grade with the sheet		Signal at 20 min
17:18–17:27	—	**Discussion** to improve the proposal (prompts below)	Plenary	See prompts below	
17:27–17:30	14–16	Session 10 on 29.10, 14:15, room 3.B47 (KW pp. 77–78, Mahoney 2000; Kurz & Schärer, Thelen 2018)	Admin		
optional	17	Backup: **dialect and AfD votes**: which controls would you add, which not?	Plenary	Could the periphery *cause* the economic controls?	Skip

Discussion prompts (if the room is quiet): The authors concede the covariates are post-treatment (fn. 9): are income, wages and unemployment confounders or mediators? · Their sensitivity benchmark is unemployment: does that help if unemployment is a mediator? · What do state fixed effects buy, and what do they cost? · A self-selected online dialect quiz: where people grew up or where they live now? Why Hannover as the standard? · Counties vs. voters: what can a county-level design say about individual motives?

Grading (0–5 each, 15/30 = pass): clarity · understanding · critique quality · reasoning · feasibility · discussion. Both presenters, one grade.

Notes during class: